

A BRIEF OVERVIEW OF AI

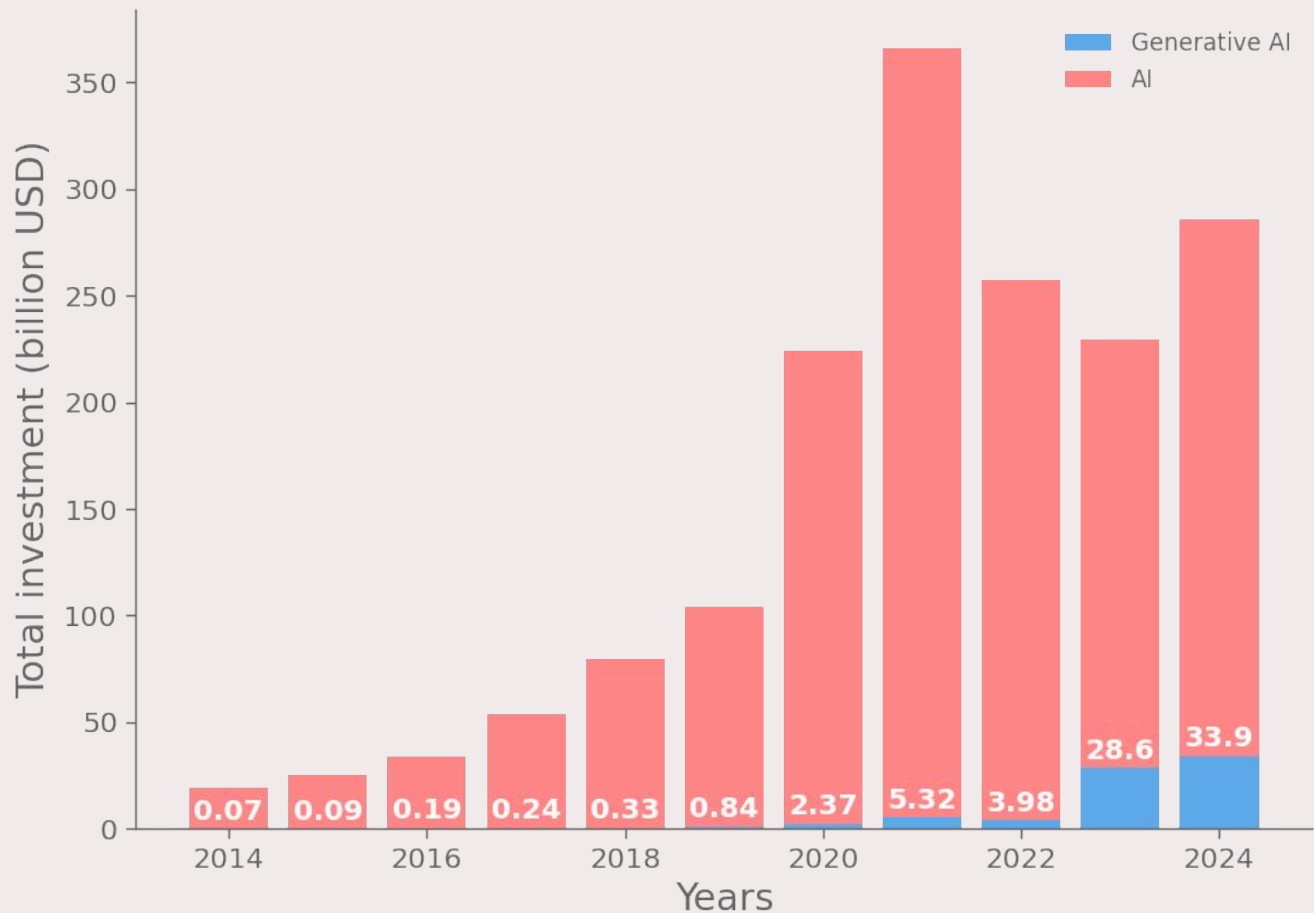
A sector by sector analysis

Giovanni Novati 68642A

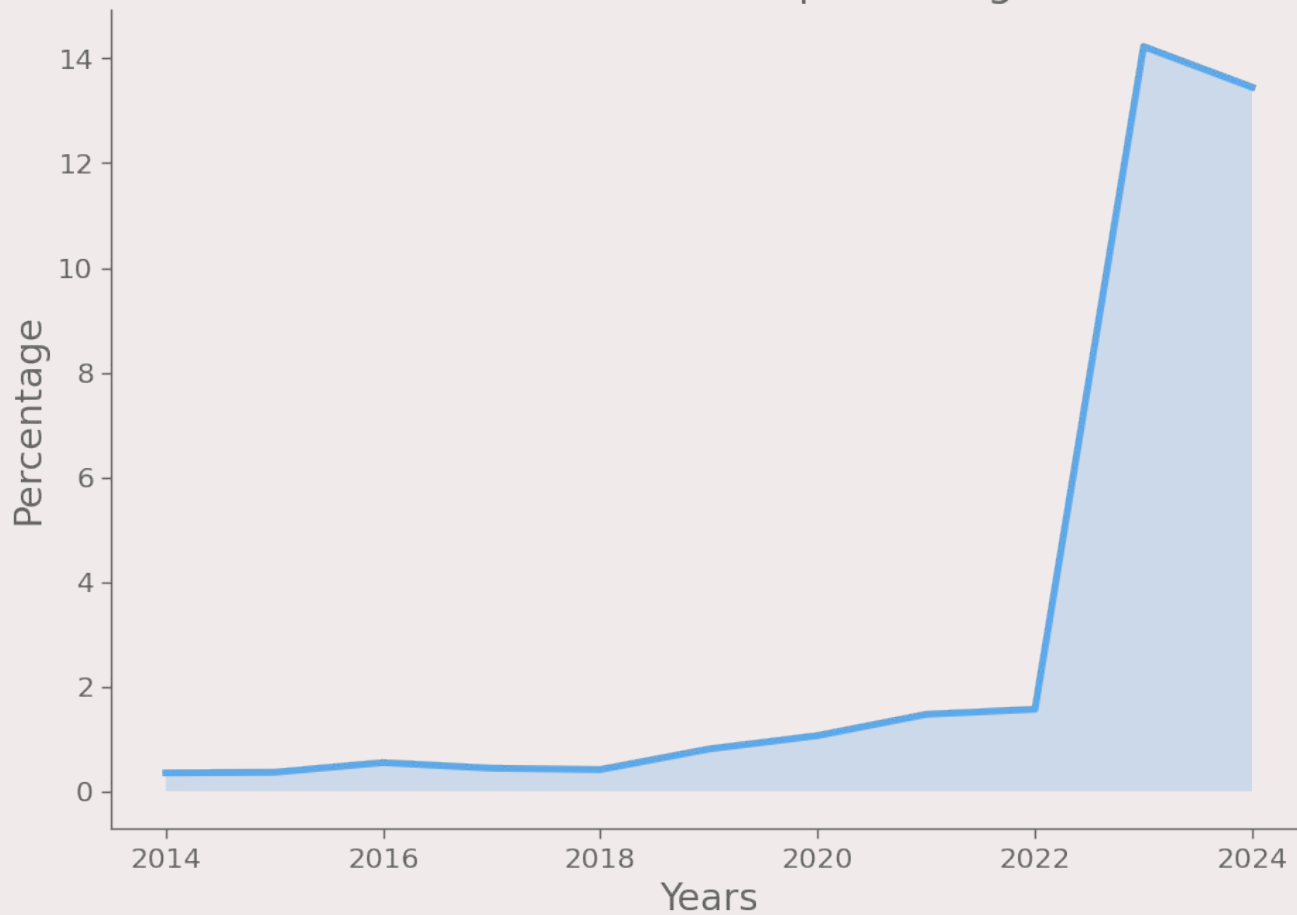
Simone Alessandro Casciaro 76678A

ECONOMY

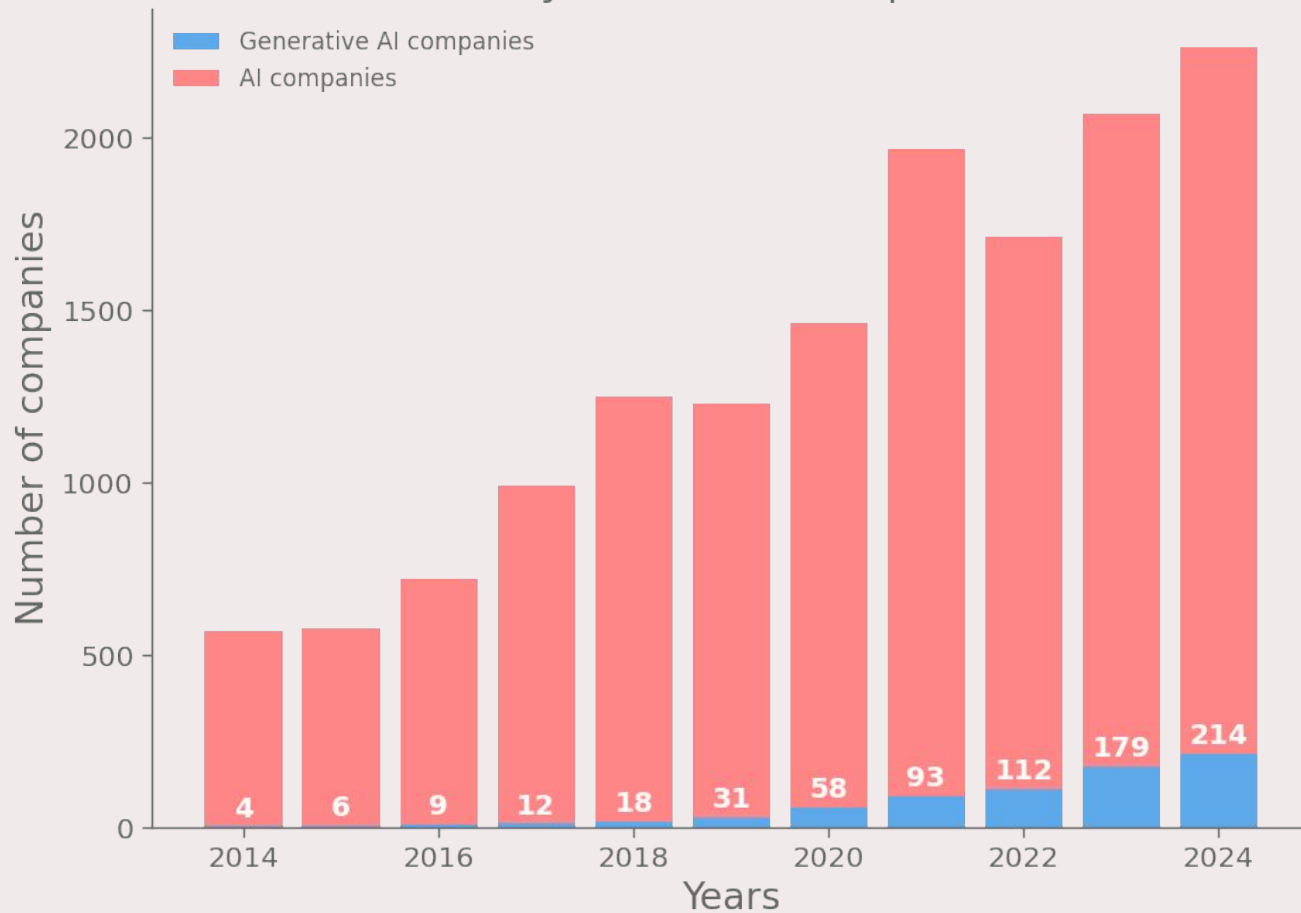
Global investment in the AI sector



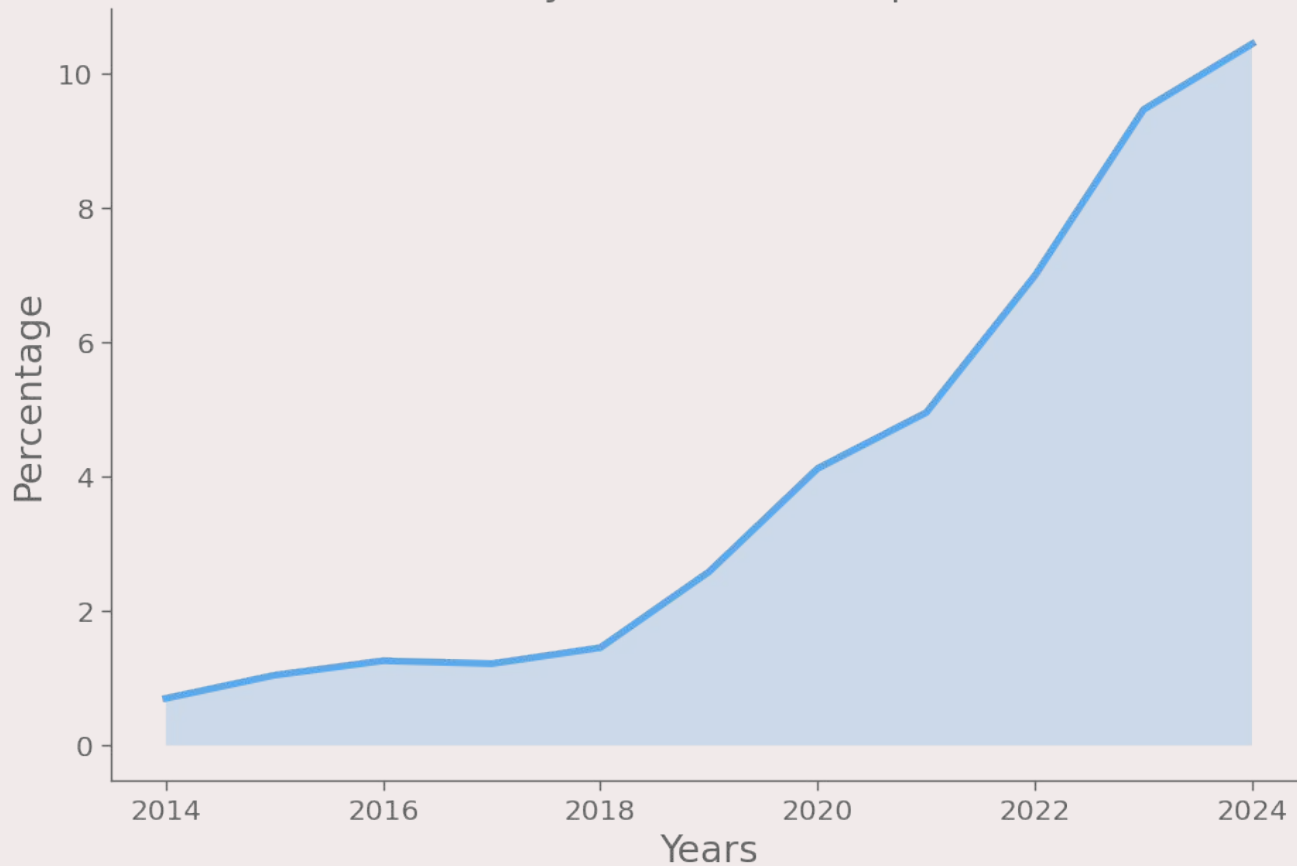
Generative AI investments as percentage of total AI



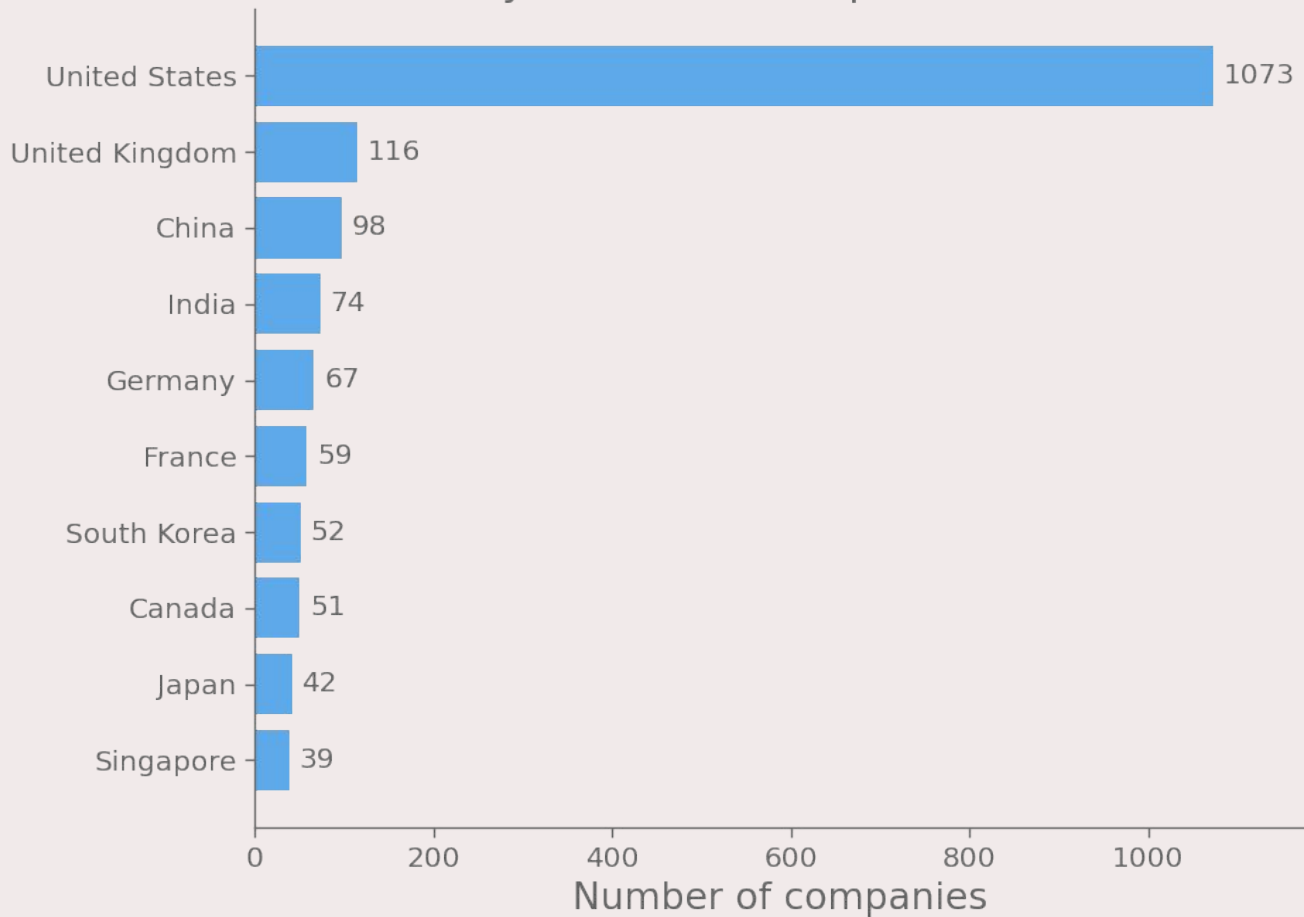
Newly funded AI companies



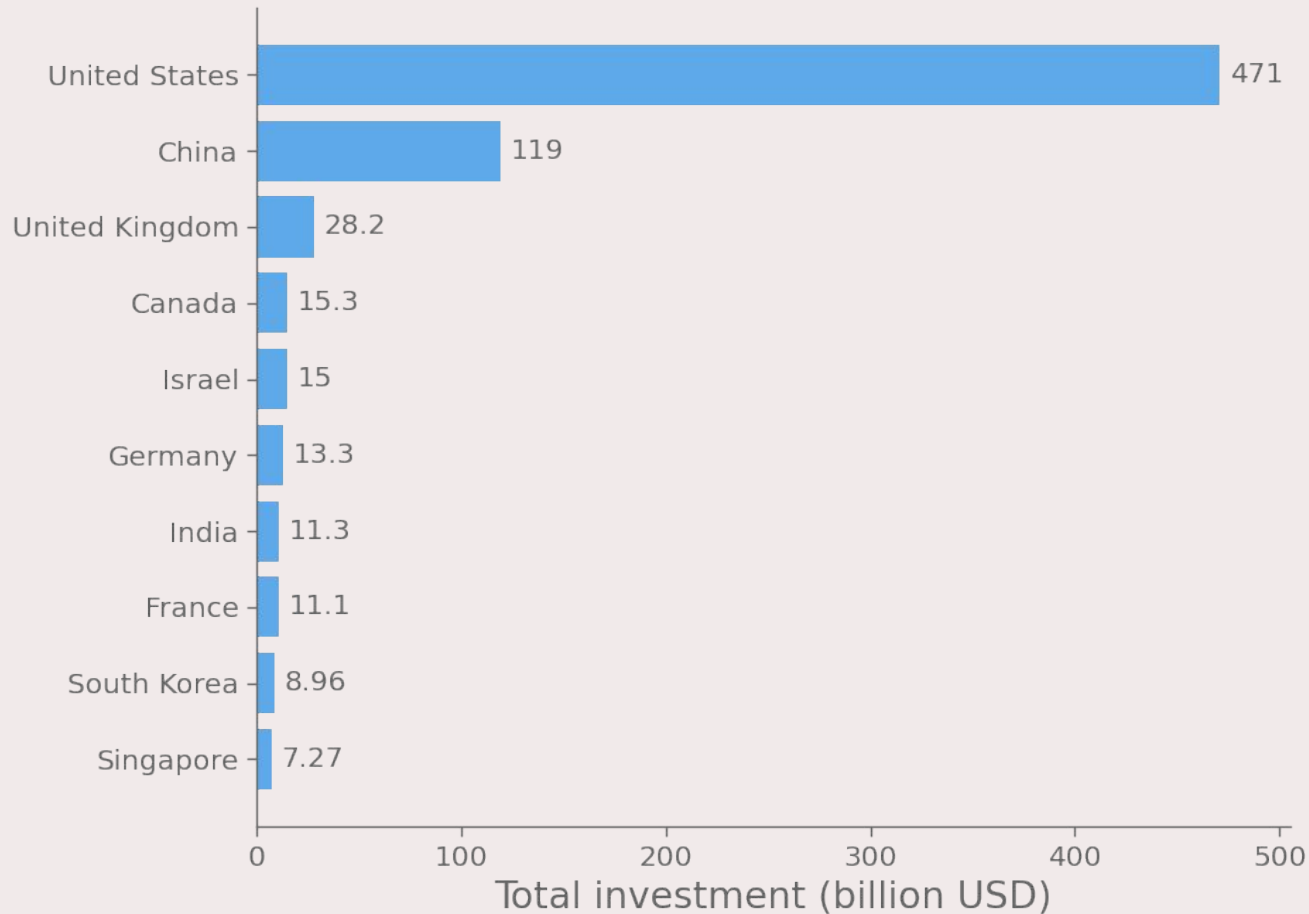
Generative AI companies as percentage
of newly funded AI companies



Newly funded AI companies in 2024

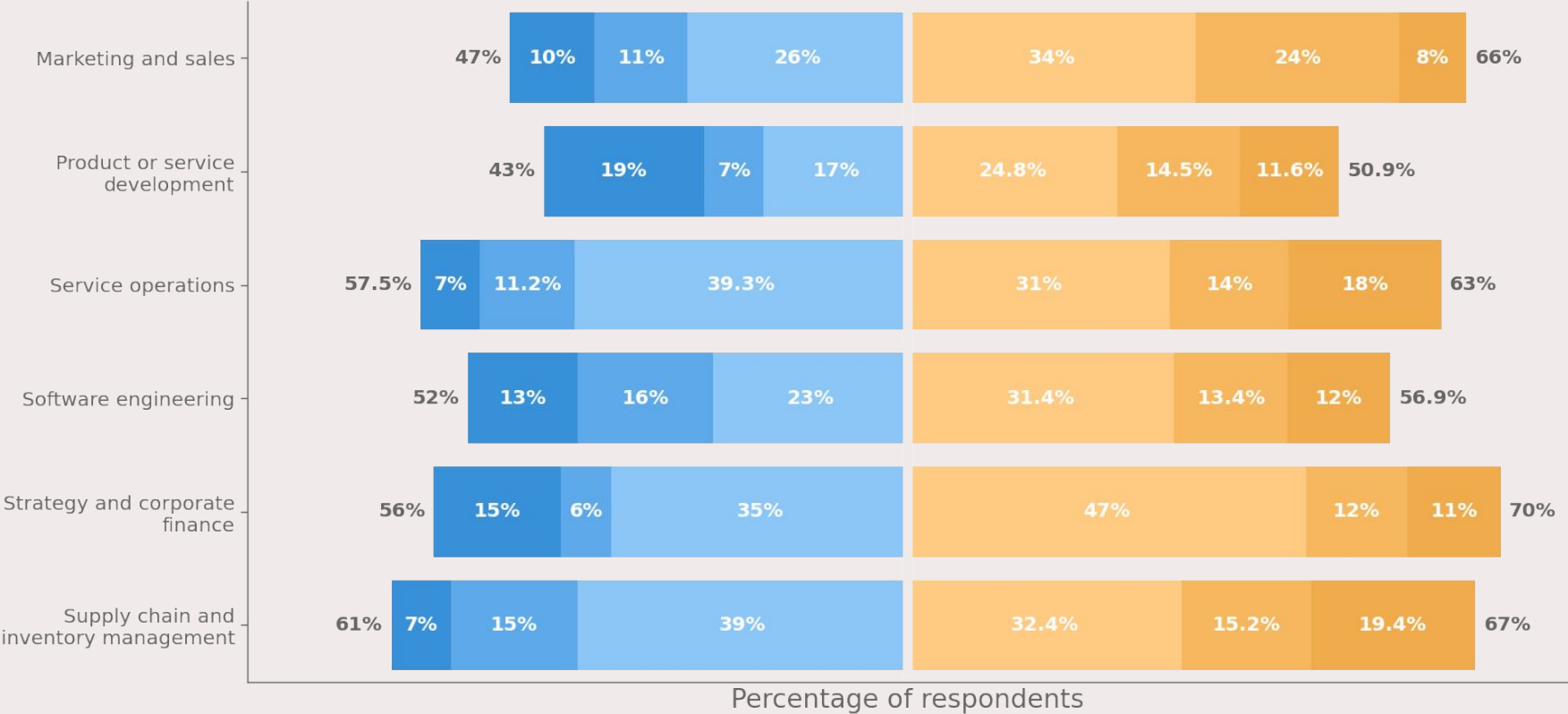


Private investments in AI from 2013 to 2024



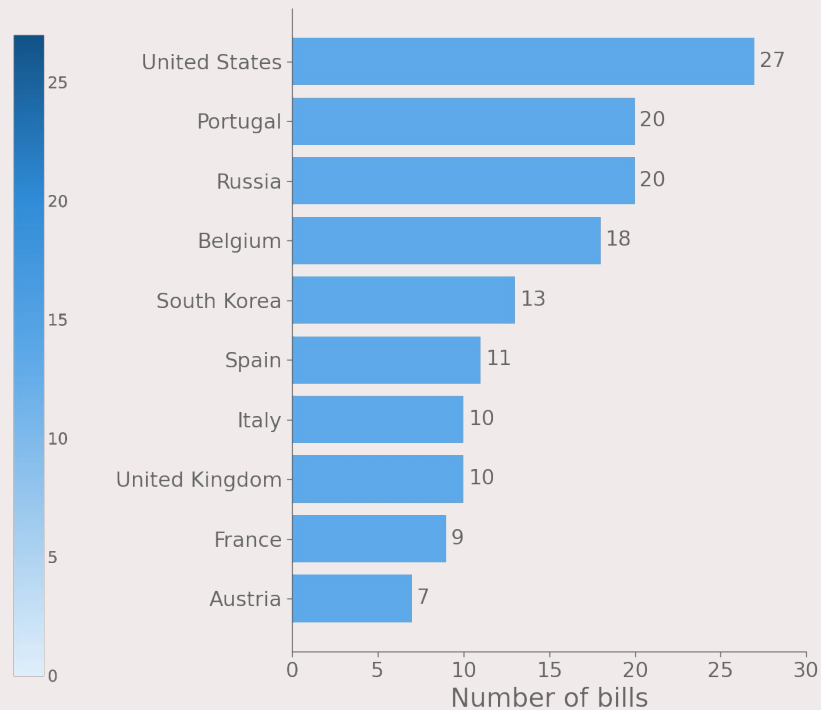
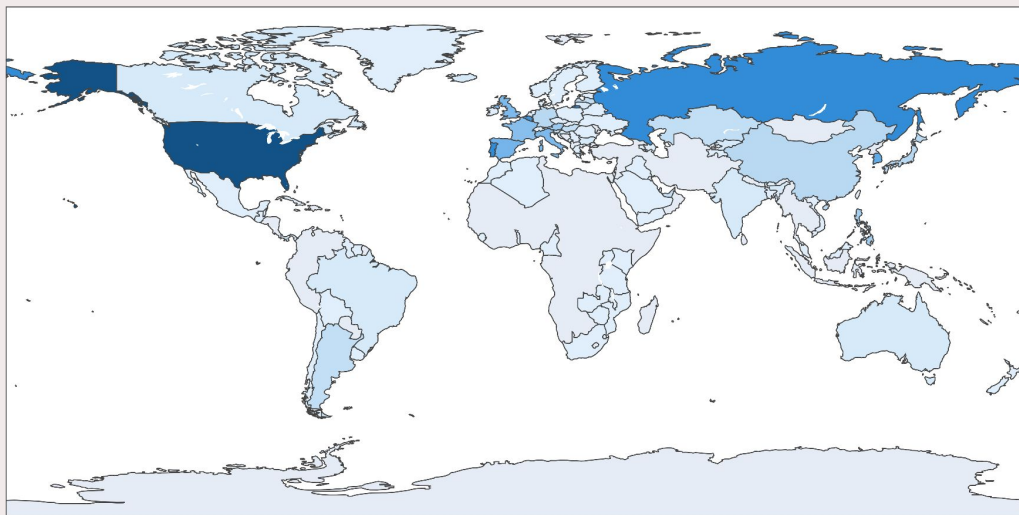
Company cost reductions and revenue gains thanks to generative AI in 2024

Decrease by ≥20% Decrease by 11-19% Decrease by ≤10% Increase by ≤5% Increase by 6-10% Increase by >10%

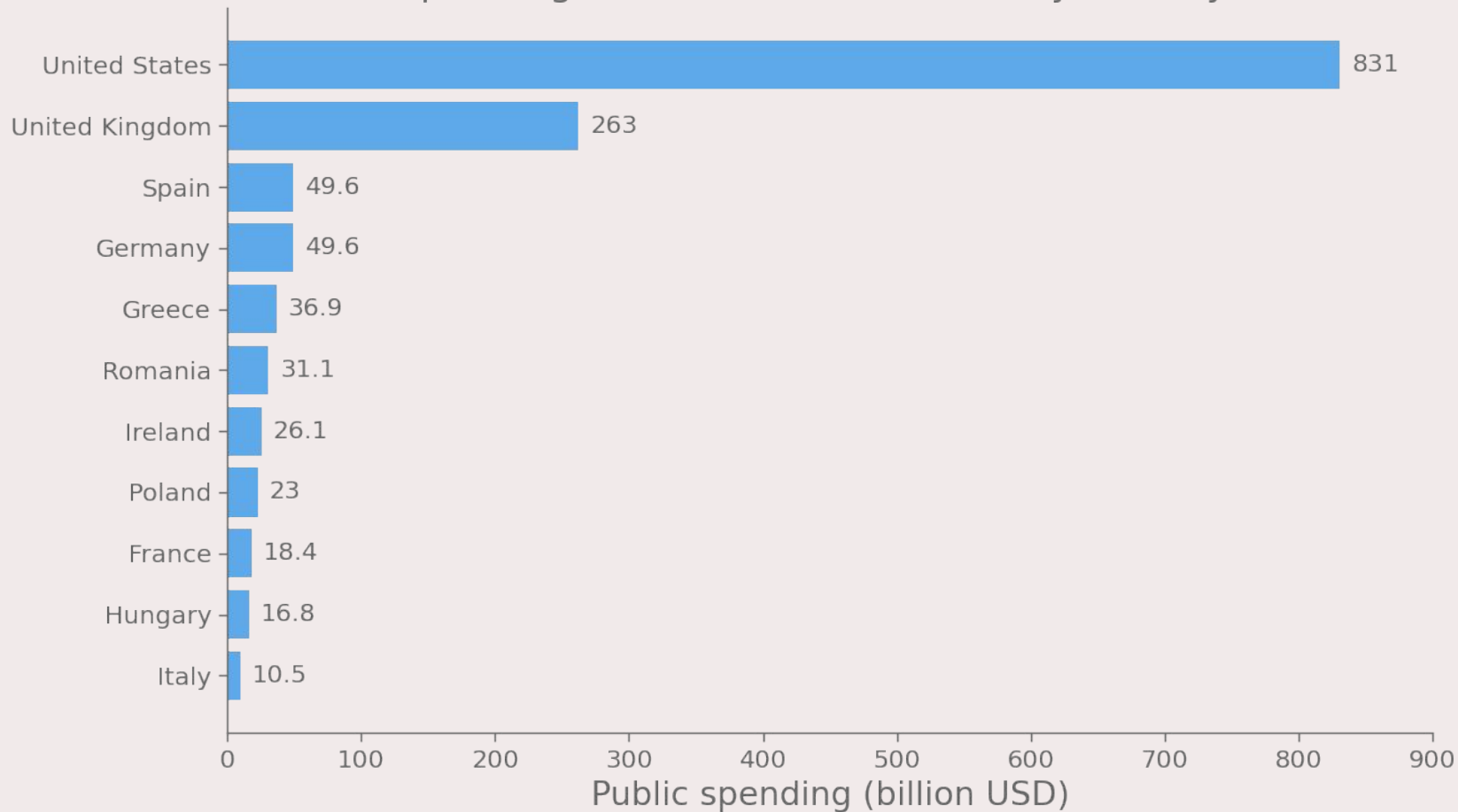


POLICY AND GOVERNANCE

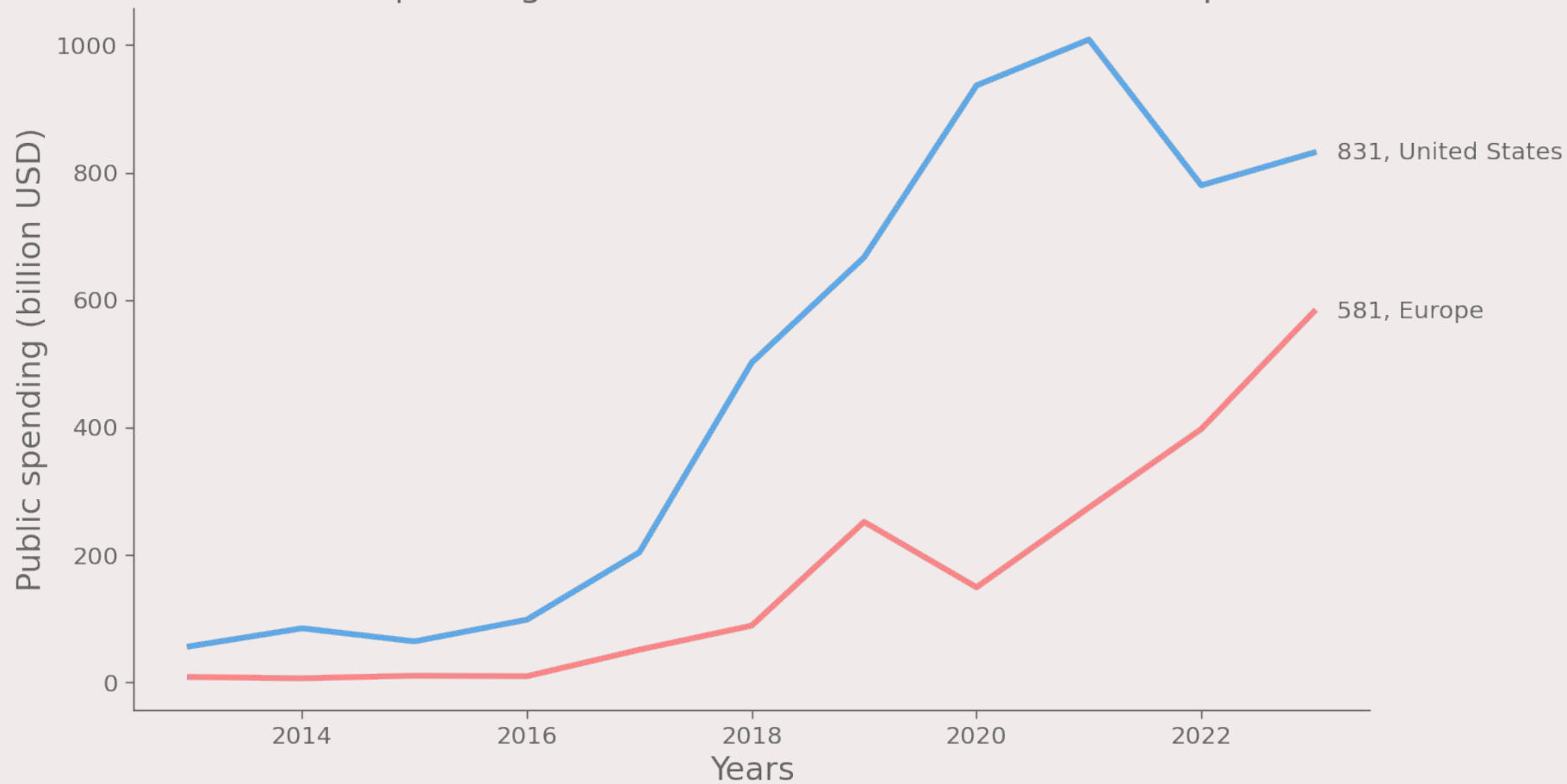
Number of AI-related bills passed into law from 2016 to 2024



Public spending on AI-related contracts by country in 2023



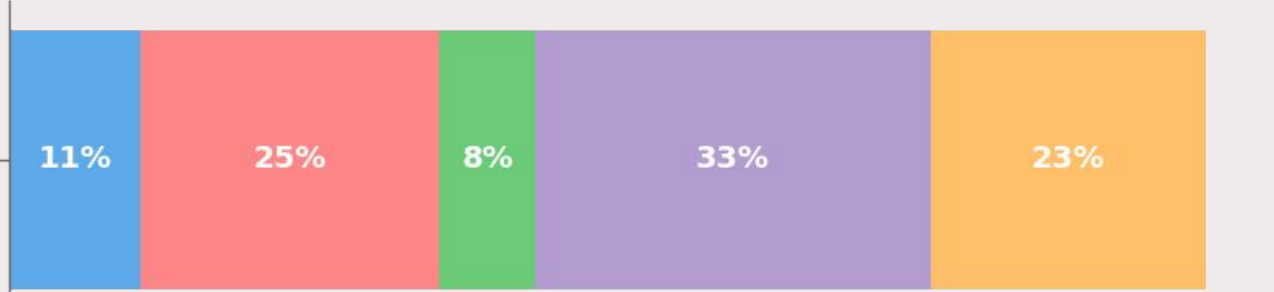
Public spending on AI-related contracts USA vs Europe



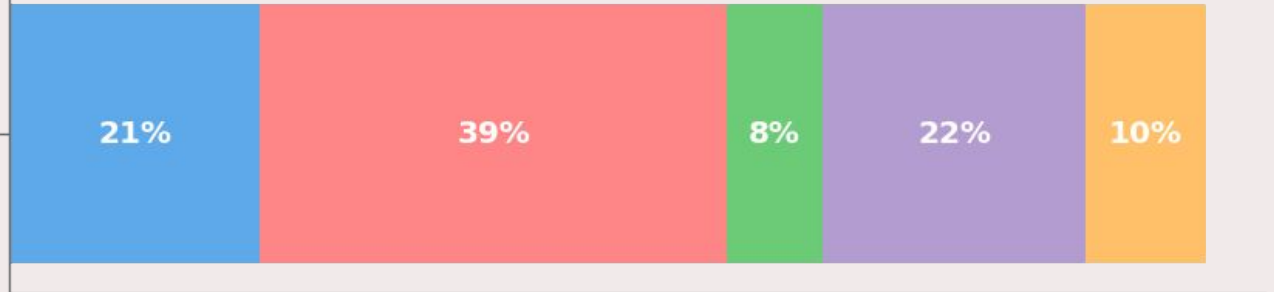
Global perceived opinion of AI on current jobs

Very likely Somewhat likely Don't Know Not very likely Not likely at all

AI will replace your current job in the next 5 years

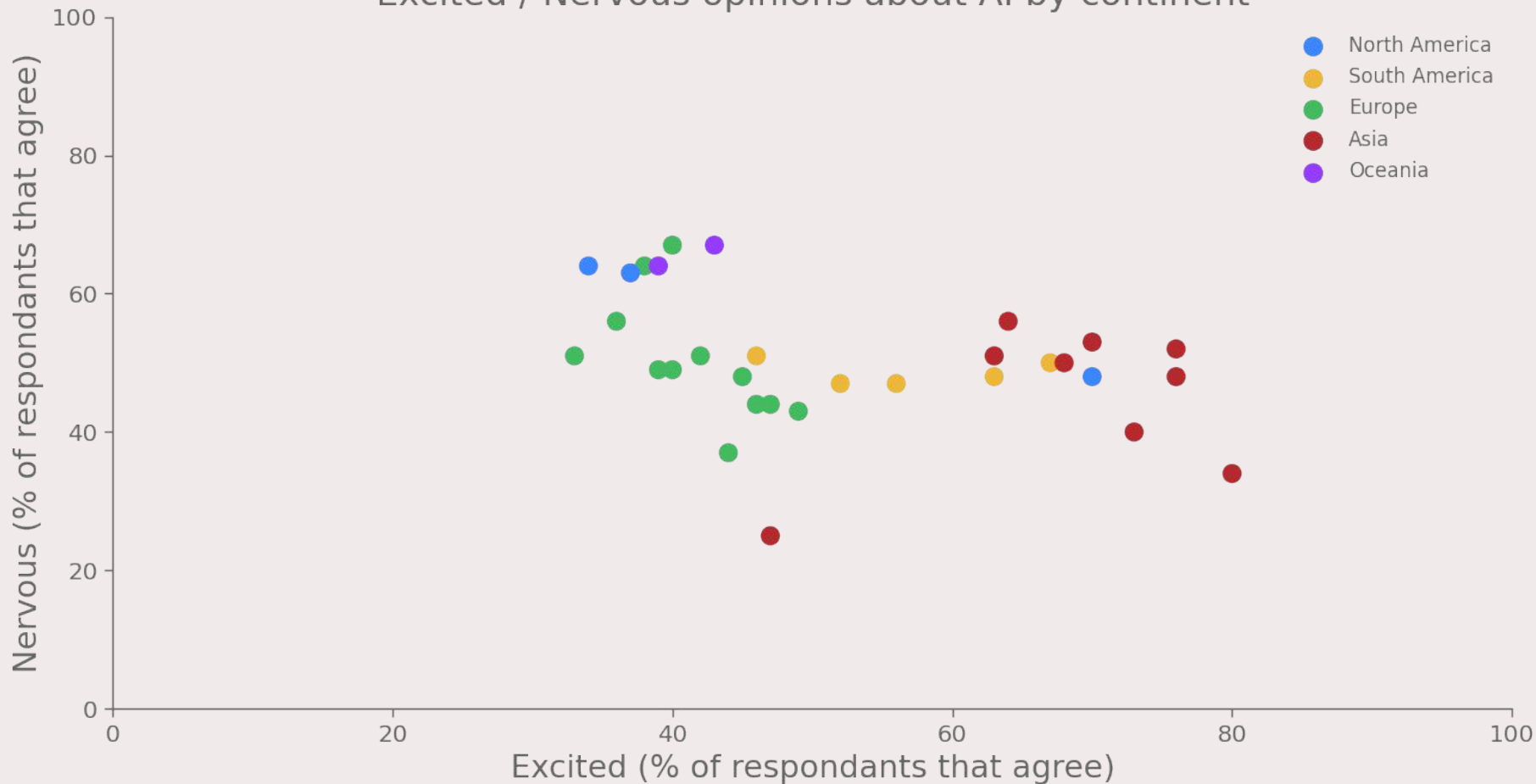


AI will change how you do your current job in the next 5 years



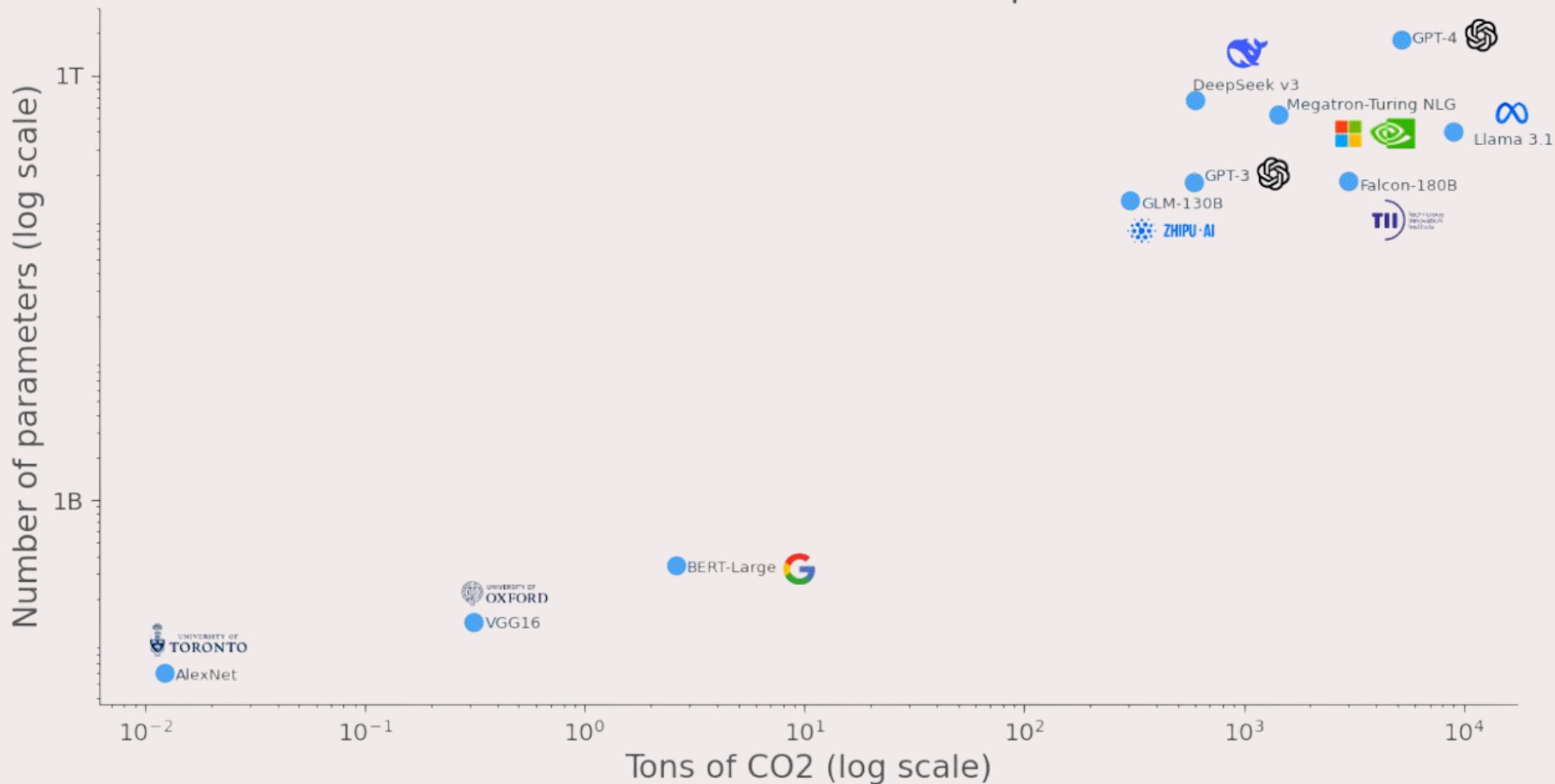
Percentage of respondents

Excited / Nervous opinions about AI by continent

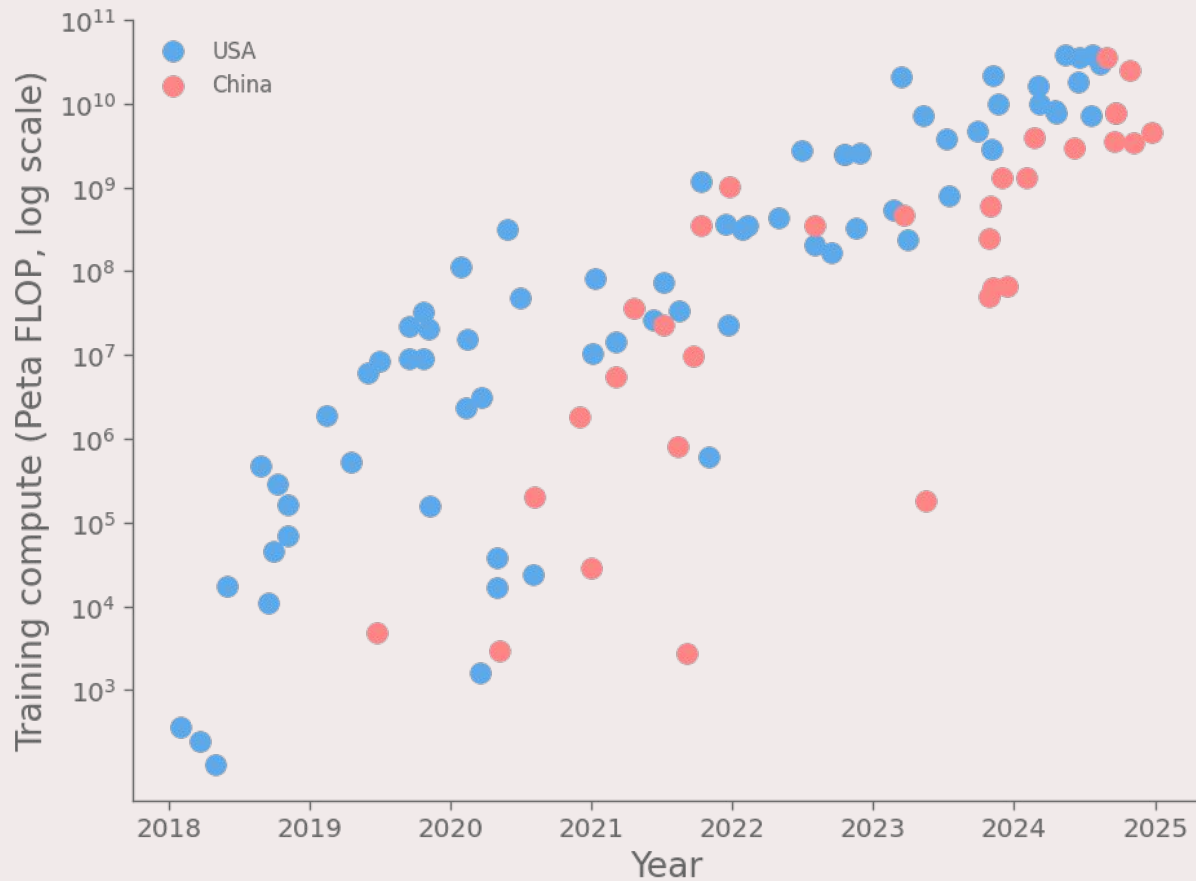


RESEARCH AND DEVELOPMENT

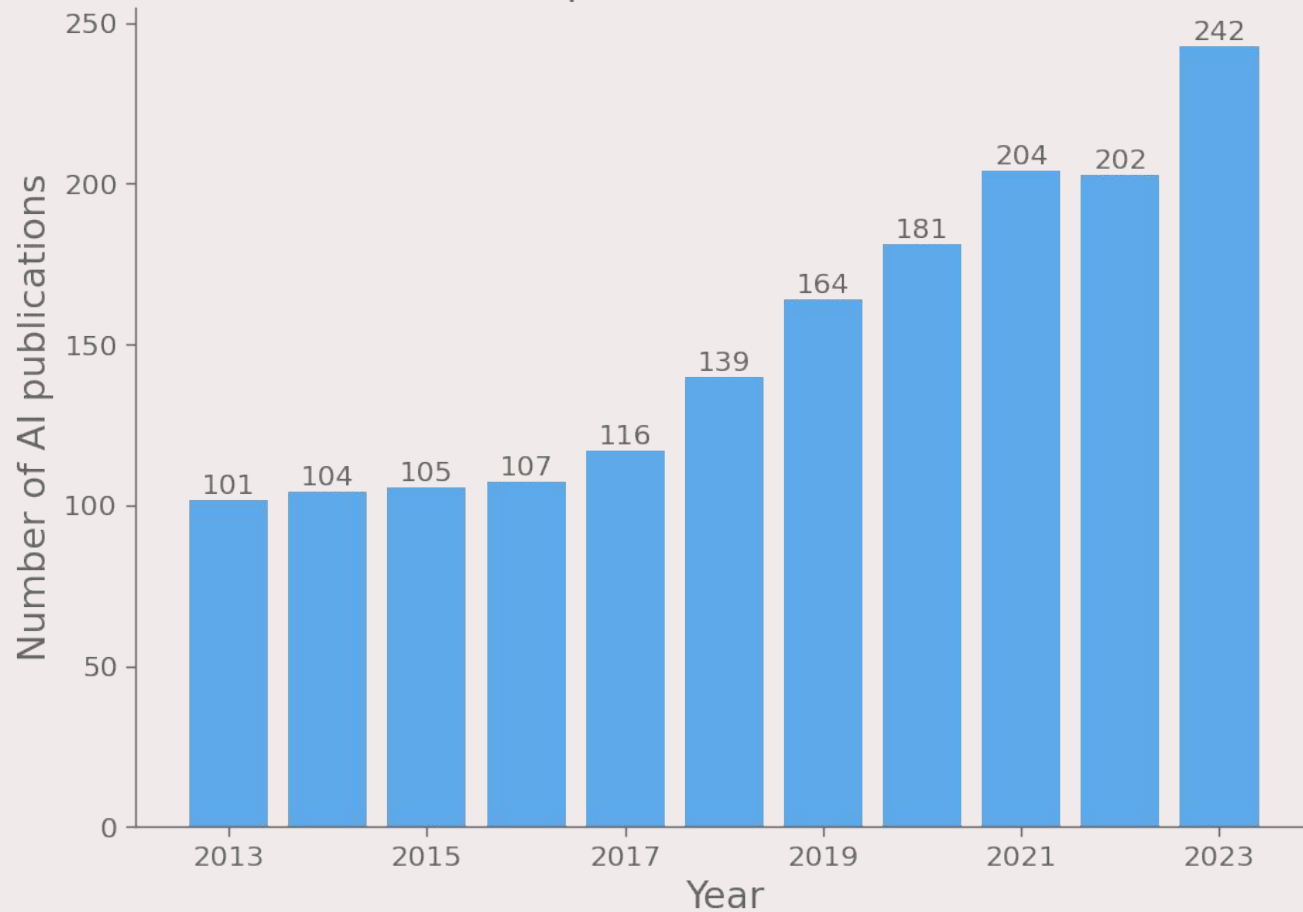
Carbon emissions and number of parameters



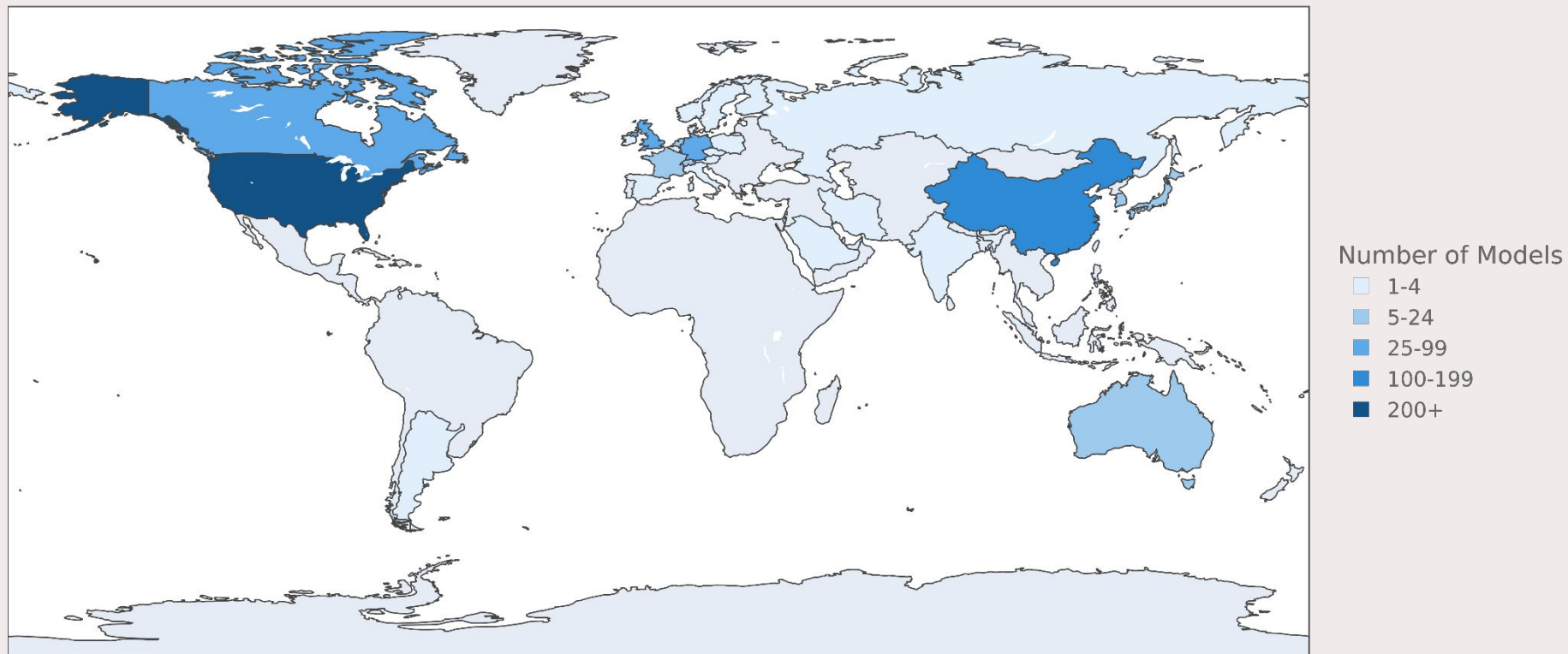
Training compute of select notable AI models in the United States and China



Number of AI publications in CS worldwide

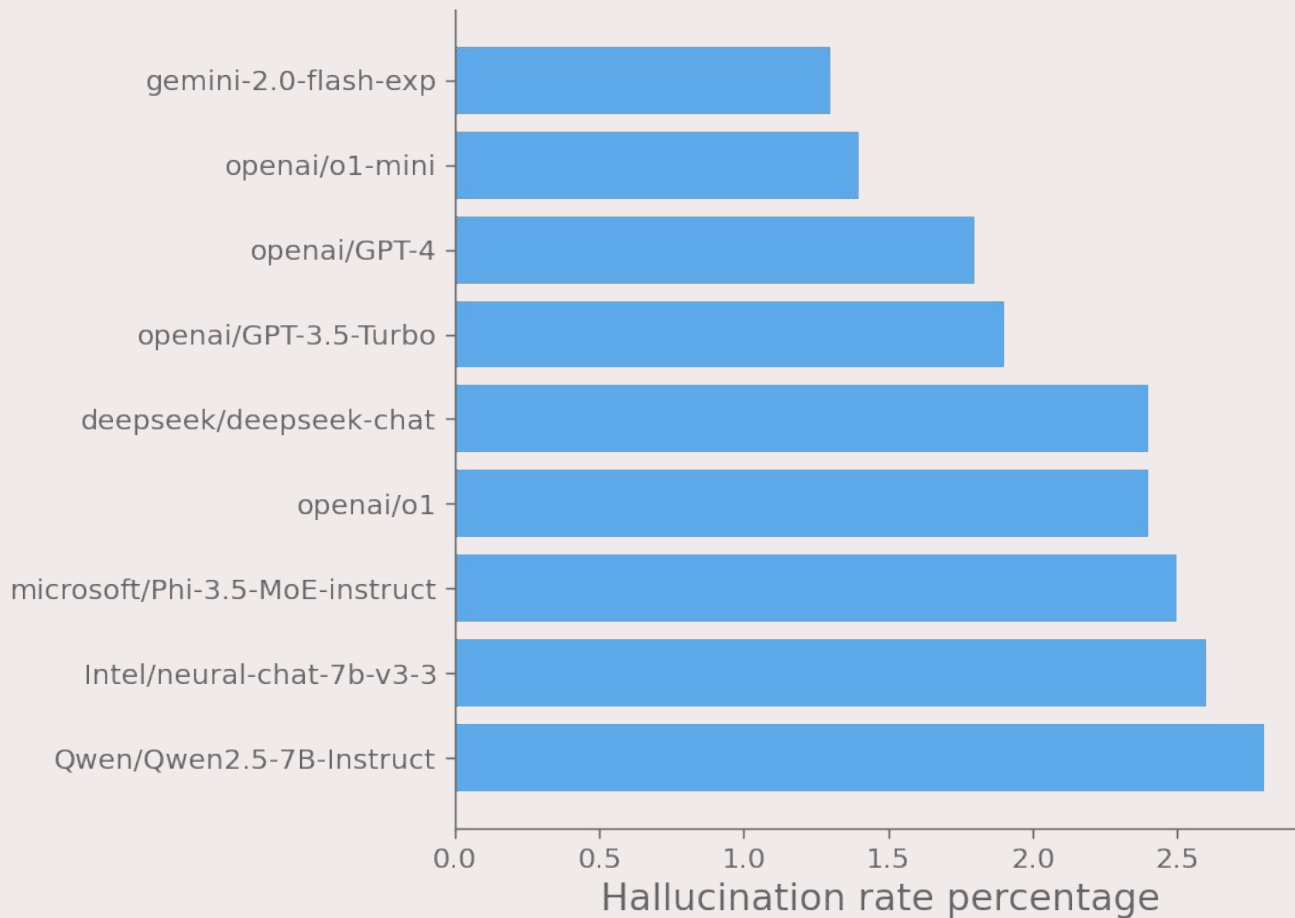


Number of Notable AI models from 2003 to 2024

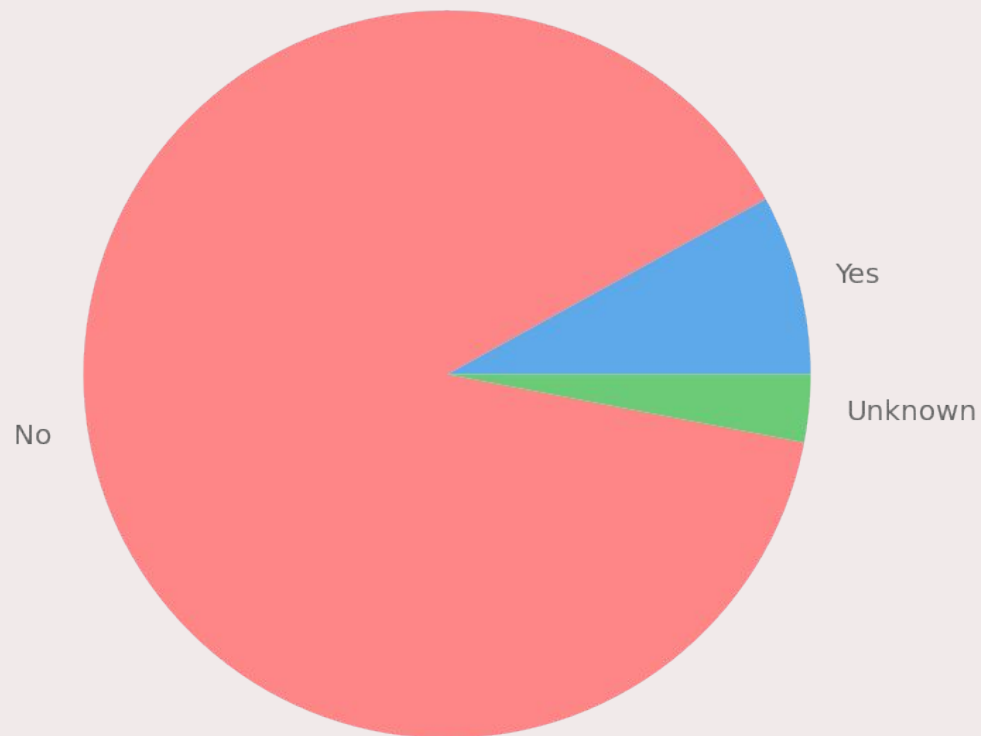


RESPONSIBLE AI

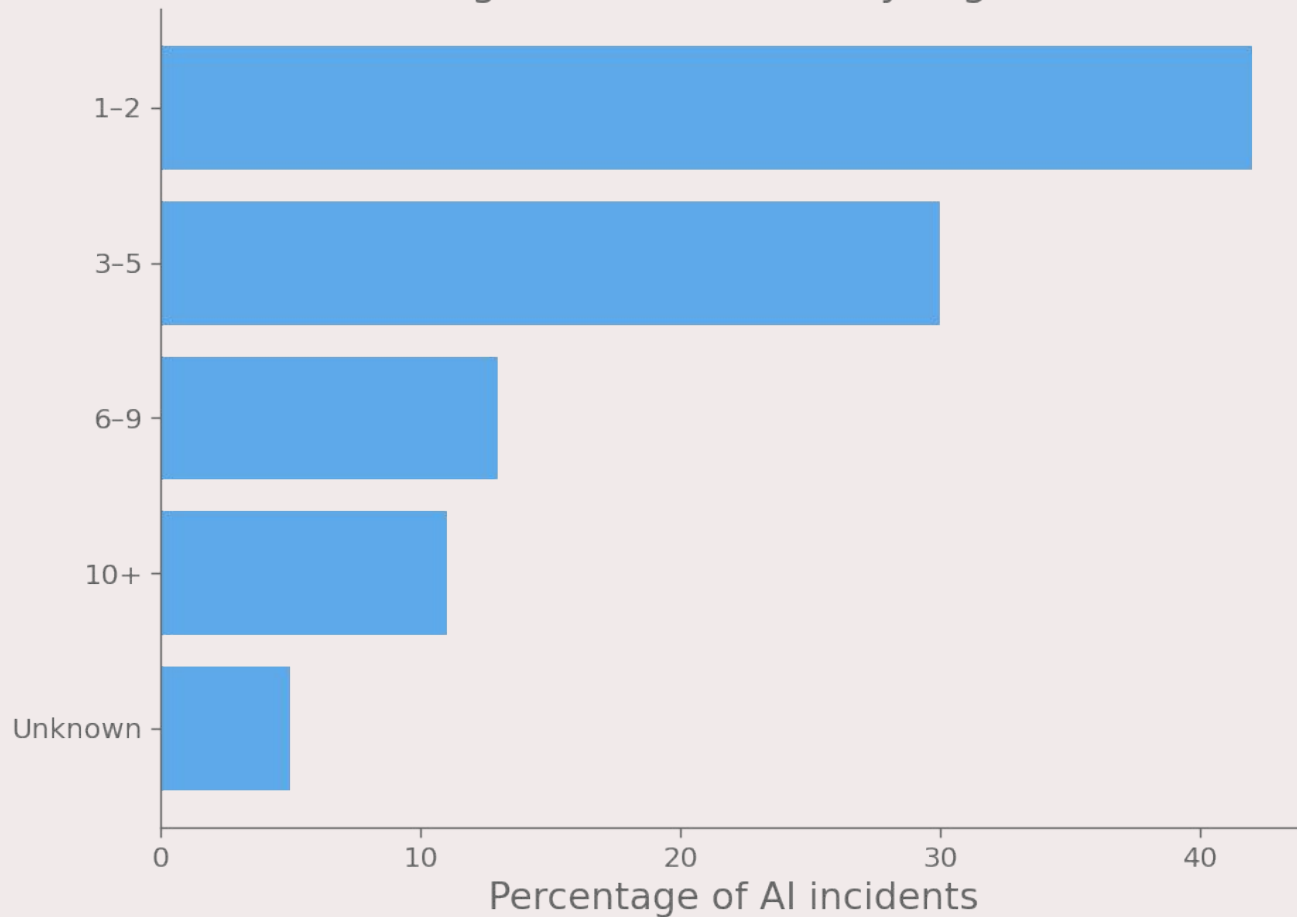
Hallucination rate of the latest models



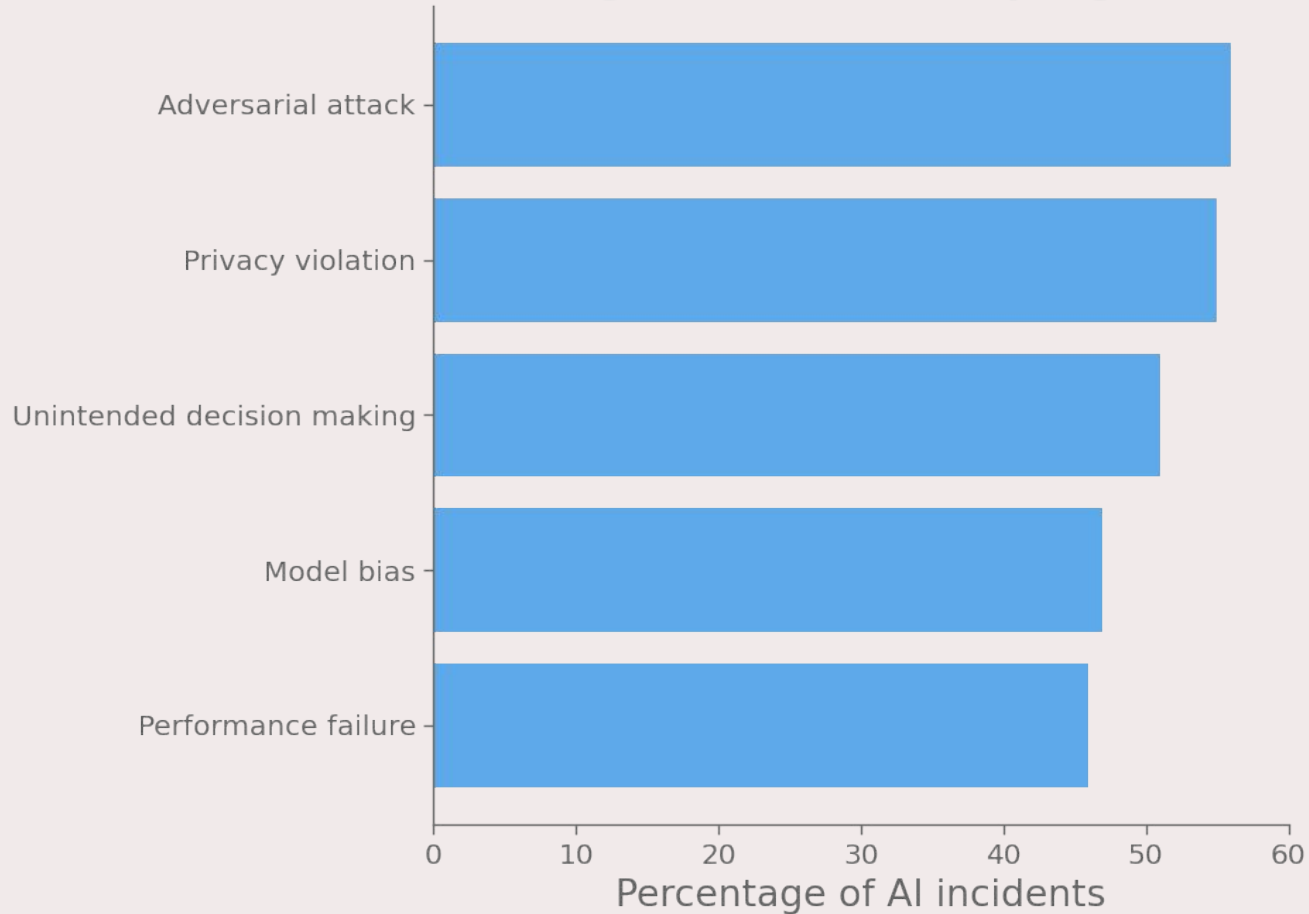
Percentage of organizations who experienced AI incidents



Percentage of AI incidents by organization

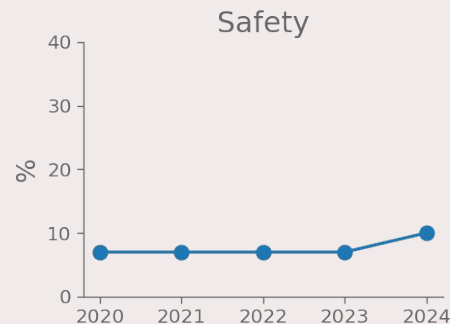
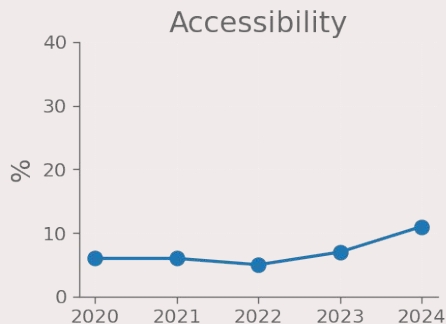
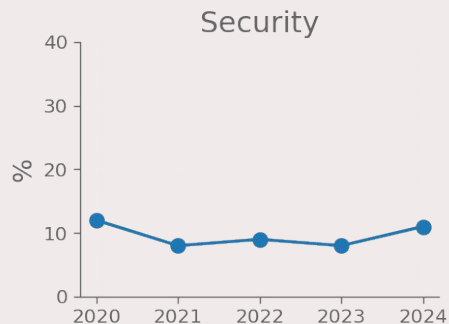
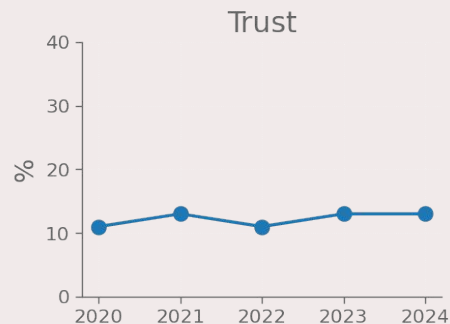
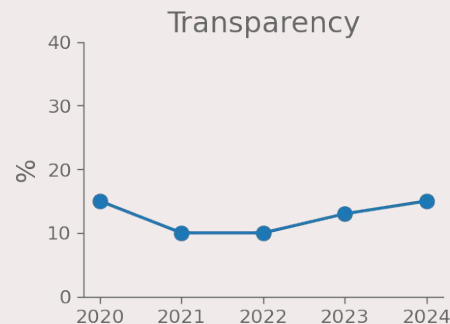
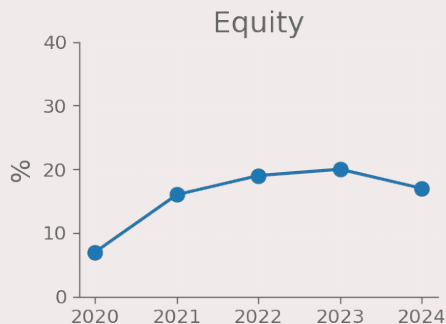
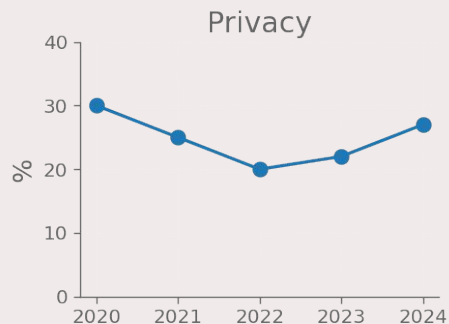
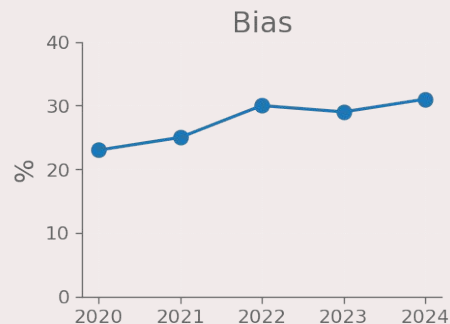


Percentage of AI incidents by organization

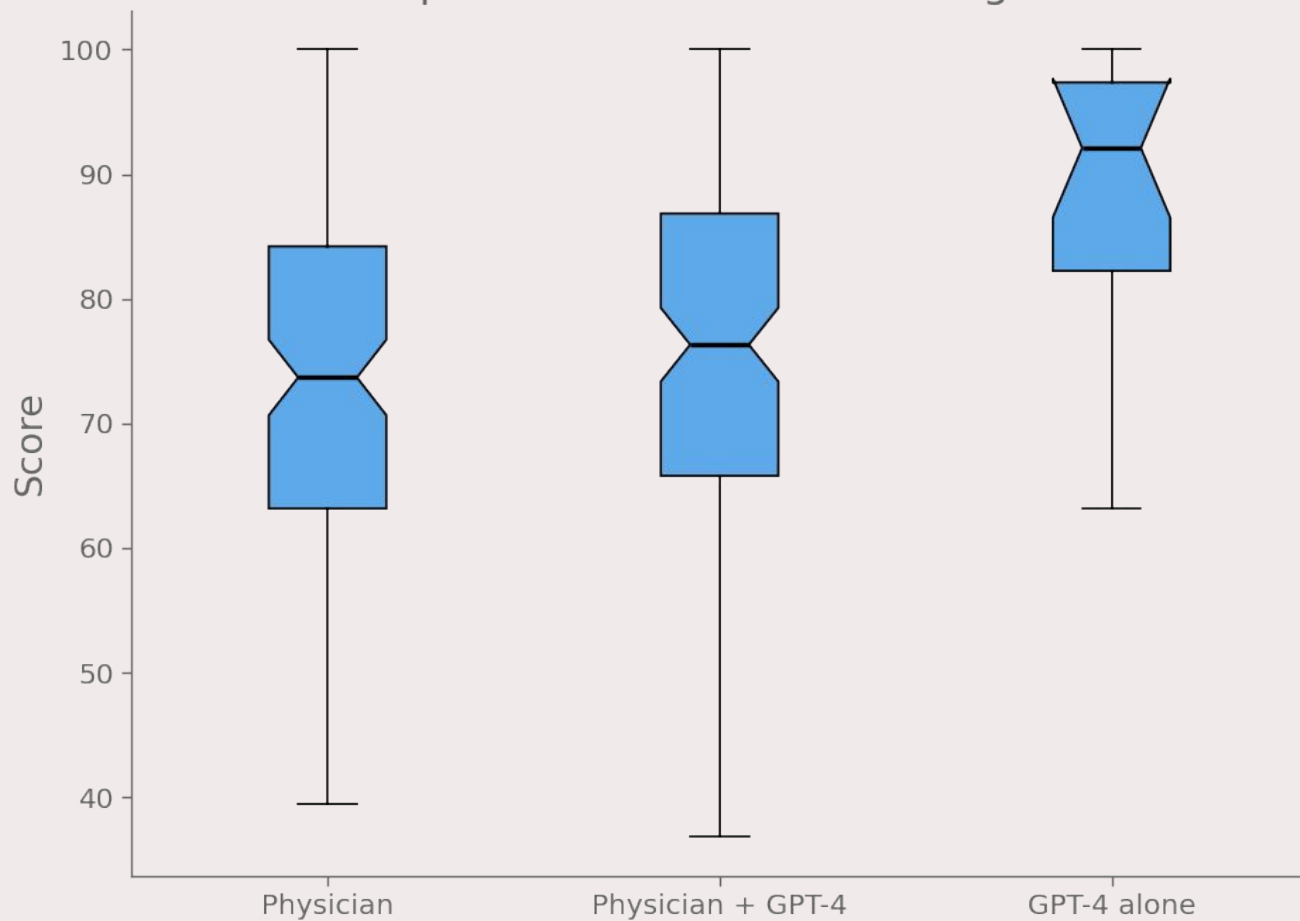


SCIENCE AND MEDICINE

Top 8 ethical concerns discussed in medical AI ethics publications

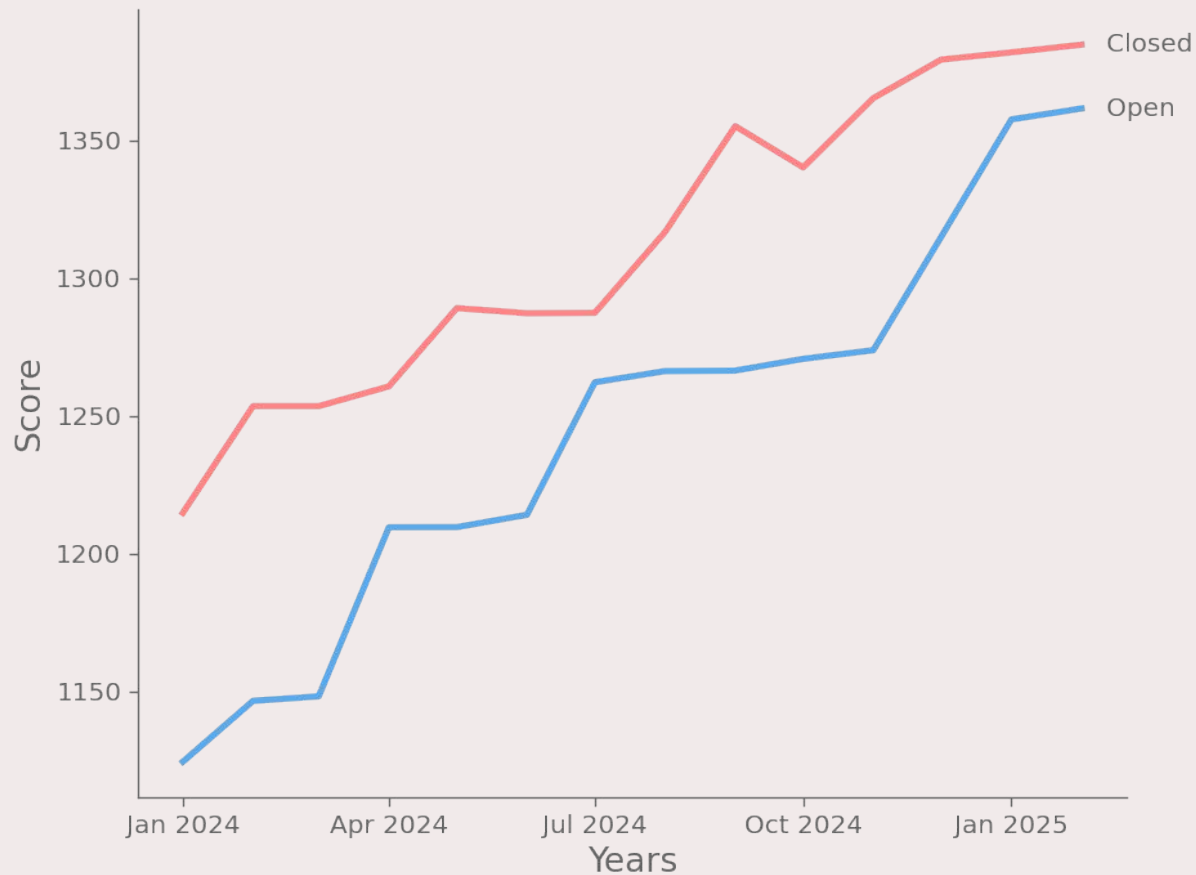


LLM performance in clinical diagnosis

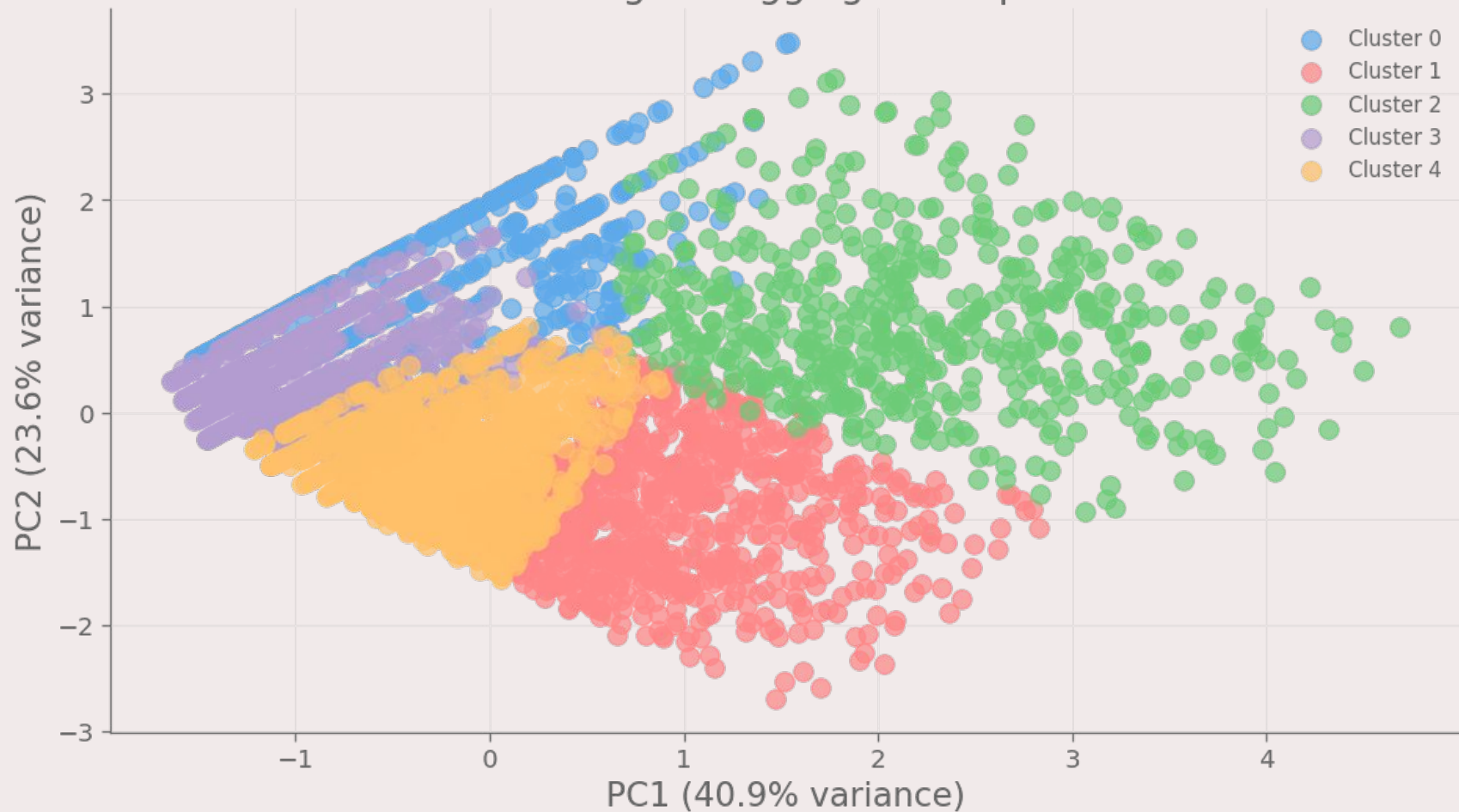


TECHNICAL PERFORMANCE

Performance of top closed/open models over years



K-Means clustering of HuggingFace top 5000 models



Cluster	Size	Avg. Downloads	Avg. Likes	Top Architecture	Top Task
0 (●)	481	6590	2	timm	image classification
1 (●)	863	3012	52	transformers	text generation
2 (●)	555	53766	30	transformers	text generation
3 (●)	1418	1655	1	transformers	text generation
4 (●)	1569	1604	7	transformers	text generation

FONTI

<https://hai.stanford.edu/ai-index/2025-ai-index-report>

<https://www.quid.com/2024-stanford-ai-index-report>

<https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai-2024>

https://ifr.org/img/worldrobotics/Press_Conference_2024.pdf

<https://www.ipsos.com/en/global-trends-2024>

<https://www.fda.gov/medical-devices/software-medical-device-samd>

<https://med.stanford.edu/raisehealth.html>

<https://epoch.ai/data/machine-learning-hardware>

<https://www.ipsos.com/en/global-trends-2024>

<https://zenodo.org/records/14801921>

GRAZIE PER L'ATTENZIONE